Study Guide, Exercises, and Sample Questions

CSE 4615: Wireless Networks

Islamic University of Technology (IUT)

Lecture 3: Enhancing Insight into Wireless Transmission Fundamentals

1. Topics Covered

- Physical layer context: role of the physical layer in wireless communication.
- Antennas: definition, types (omnidirectional vs. directional), and the transmitter chain (amplifier, oscillator, mixer, filter).
- Electromagnetic signals: time-domain and frequency-domain representations.
- Time-domain concepts: analog vs. digital signals, periodic vs. aperiodic signals, and signal parameters (amplitude, frequency, period, phase, wavelength).
- Sine wave as the carrier signal in communication; modulation of amplitude, frequency, and phase (AM, FM, PM);
- Frequency-domain concepts: fundamental frequency, spectrum, absolute bandwidth, effective bandwidth, and the relationship between bandwidth and data rate.
- Frequency spectrum of telecommunications: electromagnetic spectrum (ELF to UV), frequency bands (VLF, LF, MF, HF, VHF, UHF, SHF, EHF), wavelength–frequency relationship $\lambda = c/f$.
- Modulation types: analog modulation (AM, FM, PM), digital modulation (ASK, FSK, PSK, QAM/QPSK); modulation vs. demodulation chain.
- Combating wireless channel problems: modulation, error control coding, adaptive modulation and coding, equalization.
- Multiplexing: definition, four dimensions (space, time, frequency, code); FDM, TDM, CDM; advantages, disadvantages, and guard spaces.
- Cognitive Radio: primary users (PU), secondary users (SU), spectrum sensing, white spaces.
- Modulation vs. multiplexing: comparison of purpose, examples, and focus.
- Multiple Access: TDMA, FDMA, CDMA; reasons for widespread use of multiplexing.

2. Focus Areas

- Role of the physical layer and how it interfaces with the link layer.
- Antenna types: omnidirectional vs. directional; when and why each is used; multi-element arrays and their benefit.

- Transmitter chain: purpose of each stage (amplifier, oscillator, mixer, filter) with a numerical example (e.g., 300 MHz source $\rightarrow$ 900 MHz output).
- Difference between analog and digital signals in the time domain.
- Difference between periodic and aperiodic signals; the condition $s(t+T) = s(t)$.
- Five parameters of a sinusoidal wave: peak amplitude (A), frequency (f), period ($T = 1/f$), phase (ϕ), and wavelength (λ).
- Spectrum, absolute bandwidth, and effective bandwidth with a concrete numerical example.
- Relationship between bandwidth and data rate (more bandwidth $\Rightarrow$ higher capacity; practical costs and limitations in wireless).
- Frequency spectrum of telecommunications: locating key technologies (AM radio, FM radio, cellular, Wi-Fi, satellite, radar) on the EM spectrum.
- Wavelength–frequency relationship $\lambda = c/f$ and its relevance to antenna size.
- ISM bands: what they are, who uses them (WLANs, WPANs, IoT), and the conditions of unlicensed use.
- Analog modulation: AM, FM, PM – which parameter of the carrier is changed in each.
- Digital modulation (Shift Keying): ASK, FSK, PSK – characteristics, bandwidth needs, and robustness; QAM/QPSK concept.
- Modulation vs. multiplexing: clear distinction between the two (signal transformation vs. signal sharing).
- Multiplexing in four dimensions (space, time, frequency, code) and the corresponding multiple access schemes (SDMA, TDMA, FDMA, CDMA).
- FDM: advantages (no coordination needed, works for analog) and disadvantages (guard spaces, inflexible, wasted bandwidth).
- TDM: advantages (one carrier at a time, high throughput) and disadvantages (precise synchronization needed).
- CDM: unique codes per channel, bandwidth efficient, good interference resistance, lower per-user data rates.
- Cognitive Radio: primary vs. secondary users, spectrum sensing, white spaces; link to time/frequency/space multiplexing.
- Combating wireless channel problems: modulation choice, error coding, adaptive modulation and coding (AMC), and equalization.
- Reasons for widespread use of multiplexing: cost per kbps decreases with higher data rate; modest per-device data-rate needs.

3. How to Study These Topics

1. Sketch the transmitter chain from memory (source → amplifier → oscillator → mixer → filter → amplifier → antenna) and annotate the signal frequency at each stage using the 300 MHz → 900 MHz example from the lecture.

2. Construct a comparison table of ASK, FSK, and PSK covering bandwidth requirement, robustness to interference, and complexity.

3. Draw the four-quadrant multiplexing cube (space, time, frequency, code) and fill in one real access scheme per dimension.

4. Practice sketching FDM and TDM diagrams (as in Figure 2.12) and annotate the advantages and disadvantages of each alongside the diagram.

5. Work through the bandwidth example: given a signal spanning 100–1000 Hz with 90% energy between 300–700 Hz, compute both absolute and effective bandwidth.

6. Prepare one-page handwritten summaries of:
   - All signal parameters with definitions and units.
   - Analog vs. digital modulation types with one real-world use case each.
   - FDM vs. TDM vs. CDM comparison.

7. Clearly distinguish modulation from multiplexing using the comparison table from the lecture and create your own example for each.

4. Exercises

4.1. Antennas and the Transmitter Chain

1. Draw and label the transmitter chain. State the function of each stage.

2. Suppose the source signal is at 200 MHz and you need to transmit at 1.8 GHz. Describe the oscillator frequency required and the role of the mixer in producing the output frequency.

3. What is a multi-element antenna array? State one advantage over a single antenna.

4. Compare omnidirectional and directional antennas. Give one application domain for each.

4.2. Time-Domain Signal Concepts

5. Define the following terms and give the SI unit for each: peak amplitude, frequency, period, phase, wavelength.

6. Distinguish between a periodic and an aperiodic signal. Give one example of each in a wireless communication context.

7. A signal has frequency 500 Hz. Compute its period and wavelength in air (use $c \approx 3 \times 10^8$ m/s).

4.3. Frequency-Domain Concepts and Bandwidth

8. A signal spans from 200 Hz to 2000 Hz. However, 85% of its energy is concentrated between 500 Hz and 1200 Hz. Compute:
   - The absolute bandwidth.
   - The effective bandwidth.
9. Explain the trade-off: "More bandwidth ⇒ higher data rate, but at a cost." Give two costs specific to wireless systems.
10. Compare 4G LTE (≈20 MHz bandwidth) and 5G mmWave (≥400 MHz bandwidth) in terms of data rate potential and practical challenges.

4.4. Frequency Spectrum and Regulation

11. Compute the wavelength of a signal transmitted at 2.4 GHz (Wi-Fi) and 28 GHz (5G mmWave) using $\lambda = c/f$.
12. What are ISM bands? List three wireless technologies that operate in ISM bands and state the frequency range used.
13. Explain why VHF/UHF frequencies are preferred for mobile radio, while SHF frequencies are preferred for satellite links.

4.5. Modulation

14. Sketch the ASK, FSK, and PSK waveforms for the bit sequence 1 0 1. Identify which parameter of the carrier changes in each case.
15. Compare ASK, FSK, and PSK using the attributes: bandwidth requirement, susceptibility to interference, and implementation complexity.
16. Distinguish between analog modulation and digital modulation. Give two examples of each.
17. Draw and label the full modulation/demodulation chain from digital data at the transmitter to recovered digital data at the receiver.
18. What is adaptive modulation and coding (AMC)? Give a scenario where the system would switch from a higher-order to a lower-order modulation scheme.

4.6. Multiplexing and Multiple Access

19. Name the four dimensions of multiplexing. For each dimension, name the corresponding multiple access scheme and give one wireless system that uses it.
20. Compare FDM and TDM using the following attributes: channel assignment, synchronization requirement, suitability for analog signals, and guard requirements.
21. Construct a complete comparison table of FDM, TDM, and CDM with at least five attributes.
22. Explain why the cost of multiplexing per kbps decreases as the data rate of the transmission facility increases.

4.7. Cognitive Radio

23. Define Cognitive Radio. What problem does it solve?
24. Distinguish between primary users (PU) and secondary users (SU) of the spectrum.
25. Give two real-world examples of spectrum white spaces that cognitive radio can exploit.
26. How does cognitive radio relate to time/frequency/space multiplexing?

5. Sample Questions

5.1. Short Questions

1. Define an antenna. State its function in transmission and reception.
2. What is peak amplitude? What is its SI unit?
3. State the relationship between period and frequency.
4. Define wavelength and write the formula relating it to frequency.
5. What is the fundamental frequency of a signal?
6. Define absolute bandwidth and effective bandwidth.
7. State the four types of modulation mentioned in the lecture.
8. What is ASK? State one disadvantage of ASK.
9. What is PSK? State one advantage of PSK over ASK.
10. Distinguish between modulation and multiplexing in one sentence each.
11. What is an ISM band? Give one example of a technology that uses it.
12. Define cognitive radio in two sentences.
13. State two advantages of FDM and two advantages of TDM.
14. What is the purpose of a guard space in FDM?
15. State two reasons for the widespread use of multiplexing.

5.2. Descriptive Questions

16. Describe the complete transmitter chain, explaining the role of each component.
17. Explain the frequency-domain view of signals. Define spectrum, fundamental frequency, absolute bandwidth, and effective bandwidth. Illustrate with an example signal.
18. Describe the electromagnetic spectrum used in telecommunications. Label the major frequency bands, their approximate ranges, and give one communication technology for each.

19. Compare analog modulation (AM, FM, PM) and digital modulation (ASK, FSK, PSK). For each scheme, state which carrier parameter is varied and give one real-world use case.

20. Explain the four dimensions of multiplexing (space, time, frequency, code) with diagrams. State the corresponding multiple access scheme and one example wireless system for each dimension.

21. Compare FDM and TDM in detail. Include their principles of operation, advantages, disadvantages, guard requirements, and suitability for different traffic types.

22. Describe the relationship between bandwidth and data rate in wireless systems. Why is bandwidth considered both an asset and a constraint?

5.3. Analysis and Numerical Problems

23. A wireless signal spans from 400 MHz to 2400 MHz. 90% of its energy lies between 800 MHz and 1800 MHz.
    - Compute the absolute bandwidth.
    - Compute the effective bandwidth.
    - Compute the wavelength at the lower and upper edges of the absolute bandwidth (use $c = 3 \times 10^8$ m/s).

24. A transmitter needs to produce a signal at 1.9 GHz. The source oscillates at 400 MHz.
    - What oscillator frequency is required?
    - Describe the role of the mixer in combining these signals.
    - Why is a filter needed after the mixer?

25. An LTE base station uses FDM with channels of 200 kHz each in a 20 MHz band.
    - How many channels can it support (ignoring guard bands)?
    - If guard bands of 100 kHz separate each channel, how many channels fit?
    - Explain why guard bands are necessary.

26. A cellular system uses TDMA with 8 time slots per frame. If the frame duration is 4.6 ms:
    - How long is each time slot?
    - How many users can share one frequency channel simultaneously?
    - What synchronization challenge does this create?

27. A cognitive radio system detects that a primary user (PU) is inactive on a 5 MHz channel at 850 MHz. A secondary user (SU) wants to use it for IoT communication.
    - Which type of multiplexing does this represent?
    - What must the SU do if the PU returns?
    - Compute the wavelength of the 850 MHz signal.

5.4. Higher-Level and Design Questions

28. A wireless system designer must choose between FDMA, TDMA, and CDMA for a new cellular network. Analyze the trade-offs and recommend one scheme for a dense urban environment with 500 simultaneous users per cell. Justify your recommendation.

29. A Wi-Fi system operates at 2.4 GHz and a 5G mmWave system at 28 GHz. Compare their wavelengths, expected propagation characteristics, bandwidth availability, and suitable deployment environments.

30. Cognitive radio is proposed for TV white space reuse. Analyze the technical requirements for the SU to avoid interfering with PUs, and discuss the role of spectrum sensing in enabling this.